LOKESH SARAVANAN, Ph.D.

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Experimental materials scientist with 8+ years of research experience in functional-material synthesis, electrochemical R&D, surface/interface characterization, and nanoscale materials. Hands-on expertise on gas-phase and thin-film deposition, metal oxides/phosphides/chalcogenides, carbon and 2D materials, supported sub-nanometer Pt clusters, electrode evaluation, and advanced characterization. Strong record of connecting processing conditions, composition, morphology, surface chemistry, stability, and electrochemical performance through rigorous, reproducible experiments. Published in ACS Applied Materials & Interfaces, ACS Electrochemistry, Carbon, and Coordination Chemistry Reviews and co-authored in JACS.

Technical Skills

- **Materials synthesis & processing:** CVD, PVD, sputtering, PLD, gas-phase size-selected cluster deposition, hydrothermal synthesis, sol-gel, electrospinning, surface functionalization, and ion implantation.
- **Surface & interface characterization:** *In situ* and *ex situ* XPS, ISS, synchrotron photoelectron spectroscopy at NSRRC, Taiwan, STM, AFM, Raman spectroscopy, and surface-chemistry analysis.
- **Structural & microscopy techniques:** XRD, FE-SEM, aberration-corrected TEM/STEM, UV-Vis spectroscopy; phase, morphology, composition, and structure-property correlation.
- **Materials systems:** Sub-nanometer Pt clusters, metal oxides, metal phosphides, metal chalcogenides, carbon/2D materials, TiO_2@carbon nanowires, Ru-modified WP_2, and supported electrocatalysts.
- **Electrochemical R&D:** Electrochemical workstations; HER and ORR evaluation; electrode performance and stability testing with controlled gas exposure; *in situ*/*ex situ* electrochemical cell design; reproducibility analysis and experimental troubleshooting.
- **R&D workflow:** Process optimization, laboratory-built UHV instrumentation, root-cause troubleshooting, experimental documentation, multidisciplinary collaboration, laboratory management, mentoring, and technical presentations.

Work Experience

Postdoctoral Research Associate (University of Utah, Prof. Scott L. Anderson) **May 2024 - Present**

- Investigate size-selected and sub-nanometer Pt cluster catalysts on carbon-based electrode supports using gas-phase deposition, surface analysis, and electrochemical testing to connect deposition conditions, support chemistry, and catalytic activity.
- Develop oxygen-functionalized and nitrogen-implanted HOPG supports and systematically evaluate how ion implantation, controlled gas exposure, and surface functionalization influence Pt anchoring, stability, and ORR behavior.
- Design and use laboratory-built in situ and ex situ electrochemical cells and controlled experimental workflows to improve measurement reliability and enable surface-sensitive studies.
- Combine XPS, synchrotron photoelectron spectroscopy, STM, AFM, electron microscopy, Raman spectroscopy, and electrochemical measurements with DFT collaborations to establish nanoscale structure-property relationships.
- Evaluate transient and persistent effects of air exposure on supported Pt catalysts; use surface/composition and electrochemical data to identify performance changes and improve workflow reproducibility.
- Collaborate with researchers at Pacific Northwest National Laboratory (PNNL), Oak Ridge National Laboratory (ORNL), UCLA, and national user facilities; mentor junior researchers and contribute to laboratory operations.
- Research outputs include Pt-cluster anchoring/ORR published in ACS Electrochemistry (2026), effect of air exposure on catalysts for electrochemical reactions ACS Electrochemistry (2026), and collaborative JACS (2025) research on energetic Pt cluster deposition for HER.

Graduate Researcher (National Dong Hwa University, Taiwan) **August 2018 - February 2024**

- Developed functional catalysts for visible-light-assisted reactions and electrochemical hydrogen production, with emphasis on synthesis-processing-performance relationships.
- Designed and optimized Ru-modified tungsten diphosphide on carbon fiber, metal-oxide/phosphide systems, and rutile TiO_2@carbon core-shell nanowires for electrocatalytic and photoelectrochemical applications.
- Applied XRD, FE-SEM, AFM, Raman, UV-Vis, XPS, and electrochemical measurements to quantify phase formation, morphology, surface chemistry, stability, and performance trends.
- Used iterative composition and surface-modification strategies to improve HER performance and troubleshoot synthesis and measurement variability.
- Published first-author research in ACS Applied Materials & Interfaces (2024) and Carbon (2021); presented work at ACS Fall-2023 and Taiwan Physical Society meetings.

TEEP@Asia Plus Intern (National Dong Hwa University, Taiwan) **April 2018 - September 2018**

- Investigated carbon-containing rutile-phase TiO_2 nanowires for photocatalytic activity and developed experience in nanomaterial synthesis, structural characterization, and performance analysis.

Selected Earlier Materials Research & Training **2014 - 2016**

- University of Mumbai: synthesized graphene oxide by the Hummers method and evaluated synthesis approaches through literature-guided process comparison.
- Karunya University: hands-on training in XRD, SEM, and electrospinning; completed a mini-project on CZTS synthesis.
- Biolim Centre for Science and Technology: synthesized copper nanoparticles via a green route and evaluated antimicrobial/heavy-metal sensing applications.

Leadership & Scientific Communication (University of Utah) **2025 - Present**

- Currently a Communications Chair and served as a Junior Chair at Utah Postdoctoral Association; support postdoctoral outreach and scientific communication initiatives.
- Organizer, ComSciCon-ACS 2025 and 2026 science communication workshop.

Education

Ph.D. in Physics **2018 - 2024**

National Dong Hwa University, Hualien, Taiwan

Research focus: Development of novel catalysts for visible-light-assisted reactions and electrochemical hydrogen production via water splitting.

Integrated M.Sc. in Nanoscience & Technology **2012 - 2017**

Karunya Institute of Technology and Sciences, India

Research focus: Magnetoresistance at room temperature in ferromagnetic thin-film tunnel junctions.

Selected Publications

- Saravanan, L.; Wei, Y; Wang, Z; Dohnalek, Z.; Novotny, Z.; Rublev, P.; Alexandrova, A. N.; Anderson, S. L. “Tuning the Anchoring and ORR Activity of Sub-Nano Pt Clusters on HOPG via Ion Implantation and Oxygen Functionalization.” *ACS Electrochemistry, 2026.*
- Anand, P.; Saravanan, L.; Tung, C.-W.; Fu, Y.-P. “Advances in Heterostructured and Atomically Dispersed Metallic Carbon Catalysts for Efficient Oxygen Reactions in Zinc-Air Batteries.” *Coordination Chemistry Reviews*, 2026, 558, 217795.
- Saravanan, L.; Wang, Z.; Khan, R. A.; Anderson, S. L. “Transient and Persistent Effects of Air Exposure on Electrocatalytic Activity of Sub-Nano Pt Clusters on Different Electrode Supports.” *ACS Electrochemistry*, 2026, 2, 454-463.
- Masubuchi, T.; Rublev, P.; Han, Y.; Perez, P. J.; Boebinger, M.; Hwang, S.; Saravanan, L.; et al. “Highly Active Hydrogen Evolution Reaction Catalysts Formed by Energetic Ptn Cluster Deposition: Deposition Dynamics and the HER Mechanism.” *Journal of the American Chemical Society*, 2025, 147, 48192-48208.
- Saravanan, L.; Anand, P.; Fu, Y.-P.; Ma, Y.-R.; Yeh, W.-C. “Enhancing the Hydrogen Evolution Performance of Tungsten Diphosphide on Carbon Fiber through Ruthenium Modification.” *ACS Applied Materials & Interfaces*, 2024, 16, 12407-12416.
- Saravanan, L.; Patil, R. A.; Gultom, P.; et al. “Rutile-Phase TiO_2@Carbon Core-Shell Nanowires and Their Photoactivation in Visible Light Region.” *Carbon*, 2021, 181, 208-289.